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ECE2200L-01

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Experiment #2

Current-Voltage Characteristics of the PN Junction Diode

Executive Summary:

Diodes are used in conjunction with a resistor to find current by applying voltage. The Diodes can be put in forward or reverse bias which were explored in this lab. Unlike a resistor which has a linear graph, a semiconductor uses a I-V characteristic graph that shows an exponential growth when powered through the anode. When connected to the cathode, that being reversed bias current flow stops. Using and understanding diodes can help us improve our circuits we will build in the future and learning this now will help us to better understand what we are doing.

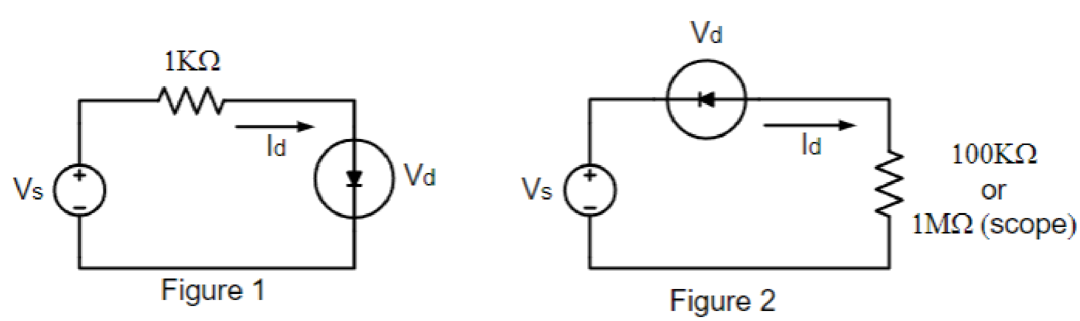
Objective:

To study the current-voltage relationship of semiconductor PN diodes, and to determine the reverse saturation current and the diode ideality factor from the I-V characteristic plot of the diodes. By finding the I-V curve we can see what the behavior of the diode is in the position of the circuit.

Pre Lab:

Acquire a school issued computer account or other access to Microsoft Excel and Spice.

Diagram:

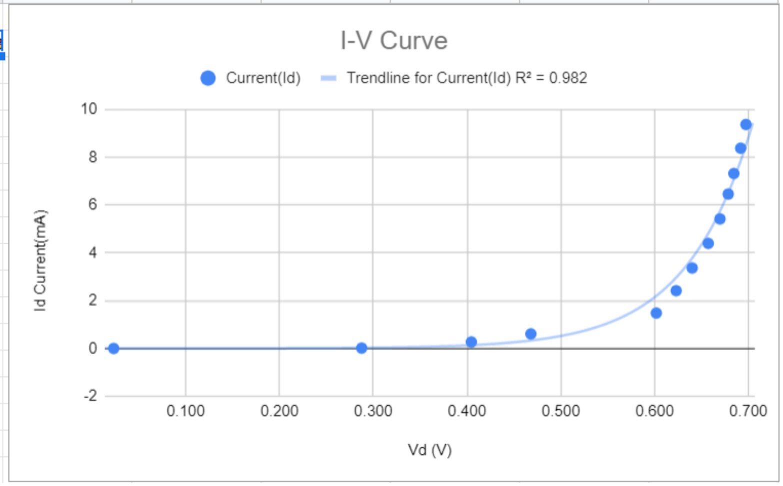


Data: Resistor used = 1kΩ, measured value = 0.993kΩ +/-7%

Part1: Forward bias diode

C2 =(A2-B2)/$D$4

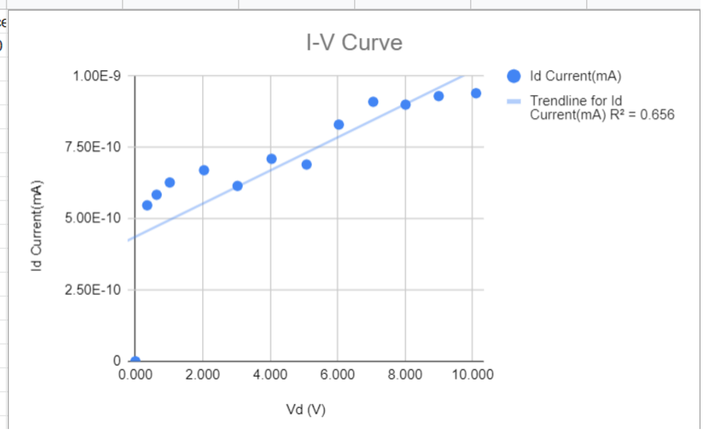
Voltage Vs (V)	Vd (V)	Current(Id)
0.02	0.023	-0.00318227593
0.30	0.288	0.01247734139
0.67	0.405	0.2673011078
1.07	0.468	0.6062114804
2.07	0.602	1.478549849
3.02	0.623	2.413897281
3.98	0.640	3.363444109
5.02	0.657	4.393554884
6.05	0.670	5.418227593
7.09	0.679	6.456696878
7.95	0.685	7.316717019
9.01	0.692	8.376837865
10.00	0.697	9.368177241



Part2: Reverse bias diode

=(B2)/$E$2

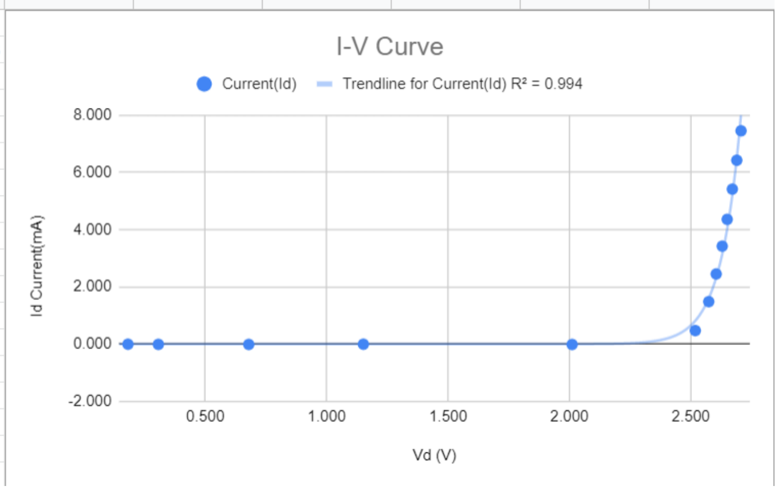
Voltage Vs (V)	VR scope (V)	Vd	Id Current(mA)	scope resistance
0.001	0.000000	0.001	0	1000000
0.352	0.000547	0.351	0.000000000547	
0.632	0.000584	0.631	0.000000000584	
1.022	0.000627	1.021	0.000000000627	
2.037	0.000670	2.036	0.00000000067	
3.031	0.000615	3.030	0.000000000615	
4.040	0.000710	4.039	0.00000000071	
5.079	0.000690	5.078	0.00000000069	
6.035	0.000830	6.034	0.00000000083	
7.055	0.000910	7.054	0.00000000091	
8.014	0.000900	8.013	0.0000000009	
8.997	0.000930	8.996	0.00000000093	
10.105	0.000940	10.104	0.00000000094	



Part3: Forward bias blue LED

=(A2-B2)/$D$2

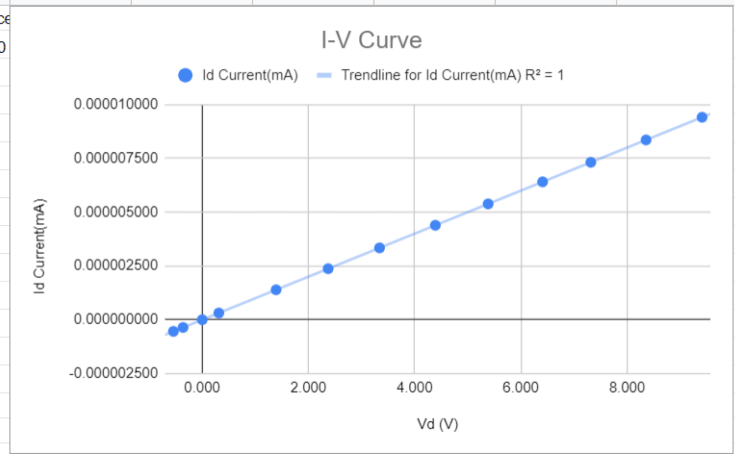
Voltage Vs (V)	Vd (V)	Current(Id)	Resistor(kohm)
0.180	0.182	-0.002	0.993
0.300	0.307	-0.007	
0.670	0.679	-0.009	
1.150	1.151	-0.001	
2.000	2.011	-0.011	
2.990	2.519	0.475	
4.050	2.574	1.486	
5.040	2.605	2.452	
6.030	2.630	3.424	
6.980	2.650	4.360	
8.050	2.671	5.417	
9.070	2.690	6.425	
10.110	2.707	7.455	
	blue led on at 3V		



Part4: Reversed bias blue LED

=(A2-B2)/$E$2

Voltage Vs (V)	VR scope (V)	Vd (V)	Id Current(mA)	scope resistance
0.110	0.650	-0.540	-0.000000540	1000000
0.312	0.670	-0.358	-0.000000358	
0.673	0.670	0.003	0.000000003	
0.985	0.670	0.315	0.000000315	
2.063	0.670	1.393	0.000001393	
3.043	0.670	2.373	0.000002373	
4.001	0.660	3.341	0.000003341	
5.051	0.660	4.391	0.000004391	
6.045	0.660	5.385	0.000005385	
7.060	0.650	6.410	0.000006410	
7.988	0.670	7.318	0.000007318	
9.018	0.660	8.358	0.000008358	
10.072	0.660	9.412	0.000009412	



Post lab:

$$Id_1 = I_s\left(e^{\frac{Vd_1}{nV_t}} - 1\right) \qquad Id_2 = I_s\left(e^{\frac{Vd_2}{nV_T}} - 1\right)$$

$$\ln\left(\frac{Id_1}{Id_2}\right) = \left(\frac{I_s\left(e^{\frac{Vd_1}{nV_t}} - 1\right)}{I_s\left(e^{\frac{Vd_2}{nV_t}} - 1\right)}\right)\ln \longrightarrow \ln\left(\frac{Id_1}{Id_2}\right) = \frac{\frac{Vd_1}{nVT}}{Vd_2/nVT}$$

$$\ln\left(\frac{Id_1}{Id_2}\right) = \frac{Vd_1 - Vd_2}{nVT} \longrightarrow n = \frac{Vd_1 - Vd_2}{VT\ln\left(\frac{Id_1}{Id_2}\right)} = \frac{0.697 - 0.692}{(0.0259)\cdot\ln\left(\frac{9.368}{6.876}\right)}$$

$$\left.\begin{array}{l} n \approx 1.725 \\ I_{s_1} = 2.351\mu A \end{array}\right\} \text{SMALL CURRENT VS OPEN CKT}$$

Analysis:

We completed four parts to this lab, two were testing forward bias and two forward bias. In the initial part of the lab, we used a diode and measured voltage across the resistor and diode. Using Ohm's law, we were able to find our current (Id). Through the I-V graph, we see that the diode once turned on as the voltage went up the line on the graph shoots up exponentially. Using the same set up by flipping the diode allowed us to go into reverse bias mode. For this an oscilloscope was required which internally has its own 1M resistance. Since it was in reversed bias, we see that the amount of current passing was very small. This is due to the current being cut off from flowing due to its reverse state. Though there was a small amount of current we still considered this 0. For the next two parts we used an LED to give a better view of what is going. Just like the diode once the LED reached activation the LED turned on which was about 3V. This means the diode is at forward bias or that it's turned on. Finally, the LED was flipped and doing so put it in reverse bias mode. As in part two, as we increased the voltage the LED stayed off. A diode can be viewed as a faucet. Turning on the voltage or faucet allows water or current to run. And turning it off won't allow the water to flow. Any values that should have been 0 could have been wrong due to leakage.

Analysis:

Using tools like excel and p-spice can not only give you a better idea of how a circuit will behave but also if it will work and both will keep you organized in the process. Using excel, the transfer function we derived can be entered and used to find the result at every number. We can then use this information to plot it as a dB vs frequency scatter plot for the circuit. This is the same result we get by AC sweep on P-SPICE after running it. From P-SPICE the given Circuit can be simulated via AC Sweep. Which we can ad a function to find Vo/vin. Furthermore, a plot was created using dB vs frequency on excel and then used to compare both. From steps two and four, they have the same shape after setting it as a log function. The become different at their drop off point where excel is about the 400 Hz point and the SPICE is around 200Hz. This can be user error by selecting more points than on the other program. P-spice and EXCEL are very powerful tools that I have used throughout my time in university and will continue to use as knowing how to use this will help with future projects and labs.